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Development of an AI-Based Weather Intelligence and Climate Decision Support Platform for the Indian Southwest Monsoon System

Part 01: Machine Learning, Deep Learning, Computer Vision,
and Time-Series Forecasting

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Abstract

The Indian Southwest Monsoon System is the primary driver of agricultural productivity, water resource availability, and socio-economic stability across the Indian subcontinent. Accurate prediction of monsoon-related meteorological variables is critical for disaster risk mitigation, agricultural planning, and climate change adaptation. This paper presents the development of an end-to-end AI-powered Weather Intelligence and Climate Decision Support Platform designed specifically for analysing and predicting the Indian Southwest Monsoon System. The platform integrates multiple machine learning approaches including traditional supervised learning models, statistical time-series forecasting, deep learning sequence models, and convolutional neural networks for satellite image analysis. Historical weather data spanning 35 years (1990–2024) was collected for eight major Indian cities from the Open-Meteo API and ERA5 Reanalysis dataset. Multiple predictive models were trained, evaluated, and compared using standard regression metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Results demonstrate that ensemble tree-based models, particularly XGBoost and LightGBM, significantly outperform both linear baseline models and standalone time-series approaches. Among all evaluated models, XGBoost achieved the best predictive performance with a Mean Absolute Error (MAE) of 2.6561 mm, a Root Mean Squared Error (RMSE) of 8.0609

mm, and an R^2 score of 0.7477 on the test dataset. LSTM-based deep learning models captured sequential dependencies in rainfall data and achieved competitive performance, demonstrating their ability to model temporal patterns without explicit feature engineering.. SHAP (SHapley Additive exPlanations) analysis reveals that recent rainfall history (lag features), cloud cover, and seasonal encoding are the most influential predictors. The complete platform is deployed as a production-ready web application using FastAPI and Streamlit, containerised with Docker, and maintained as an open-source GitHub repository available at <https://github.com/Shivyamahajan/weather-intelligence-platform>.

Keywords: Indian Southwest Monsoon, Rainfall Prediction, Machine Learning, LSTM, XGBoost, Time-Series Forecasting, Computer Vision, Weather Intelligence, Climate Decision Support

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1 Introduction

The Indian Southwest Monsoon (ISM), active from June through September, delivers approximately 70–80% of India’s annual precipitation and directly influences the livelihoods of over 600 million agricultural workers [1]. Variations in monsoon onset, intensity, duration, and spatial distribution frequently result in catastrophic floods, prolonged droughts, crop failures, and infrastructure damage, collectively causing economic losses estimated at billions of dollars annually.

Accurate forecasting of monsoon weather variables — particularly daily rainfall at city and regional levels — has therefore become a critical national priority for India’s agriculture, water resource management, disaster preparedness, and energy planning [2]. Traditional numerical weather prediction (NWP) models, while physically rigorous, require enormous computational resources and struggle to capture the complex non-linear relationships inherent in regional monsoon dynamics [3].

The emergence of machine learning (ML) and deep learning (DL) methods has opened new possibilities for data-driven weather prediction that complement and, in certain applications, surpass traditional approaches. Generative AI and large foundation models are further transforming the landscape, enabling intelligent decision support systems that combine predictive analytics with natural language reasoning.

1.1 Objectives

This research project aims to:

1. Collect, process, and analyse 35 years of historical meteorological data for eight major Indian cities
2. Develop and evaluate multiple ML, DL, and time-series forecasting models for daily rainfall prediction
3. Apply SHAP explainability analysis to understand feature contributions to model predictions
4. Develop and evaluate CNN models for satellite cloud image classification
5. Deploy all models in a production-ready web application using FastAPI and Streamlit

1.2 Contributions

The key contributions of this work are:

- A comprehensive comparative study of 12+ ML, DL, and time-series models for Indian monsoon rainfall prediction
- Feature engineering methodology specifically designed for monsoon seasonality using cyclical encoding
- SHAP-based interpretability analysis for weather prediction models
- Transfer learning adaptation of MobileNetV2 for satellite cloud image classification
- A production-ready, open-source Weather Intelligence Platform with REST API and interactive web interface

1.3 Paper Organisation

The remainder of this paper is organised as follows. Section 2 reviews relevant literature. Section 3 describes the research methodology. Section 4 presents the dataset. Section 5 describes all models implemented. Section 6 presents experimental results. Section 7 discusses findings. Section 8 concludes and identifies future work.

2 Literature Review

2.1 Traditional Weather Prediction

Traditional numerical weather prediction (NWP) models, such as those operated by the European Centre for Medium-Range Weather Forecasts (ECMWF) and the India Meteorological Department (IMD), are based on physical equations governing atmospheric dynamics [5]. While physically principled and globally applicable, these models require significant supercomputing resources and face fundamental accuracy limitations at regional and local scales due to parameterisation uncertainties.

2.2 Machine Learning for Rainfall Prediction

Numerous studies have explored ML approaches for rainfall prediction. [14] demonstrated that Random Forest models outperform linear regression approaches for monthly rainfall prediction in India. Applied artificial neural networks to Indian rainfall data, achieving improved performance over statistical baselines. Gradient boosting methods, particularly XGBoost, have shown consistent state-of-the-art performance on meteorological prediction tasks [7].

2.3 Deep Learning for Time-Series Forecasting

Long Short-Term Memory (LSTM) networks, introduced by [6], have become the dominant deep learning architecture for sequential data modelling. Their gating mechanisms address the vanishing gradient problem that limits standard recurrent neural networks. Several recent studies have applied LSTM to weather prediction tasks with competitive results [15].

2.4 Computer Vision for Satellite Analysis

Convolutional Neural Networks (CNNs) have revolutionised the analysis of satellite and weather imagery . Transfer learning approaches using pre-trained models such as ResNet, VGG, and MobileNet have demonstrated strong performance on remote sensing classification tasks even with limited training data [16].

3 Research Methodology

3.1 Overall Framework

Figure 1 illustrates the complete research methodology framework.

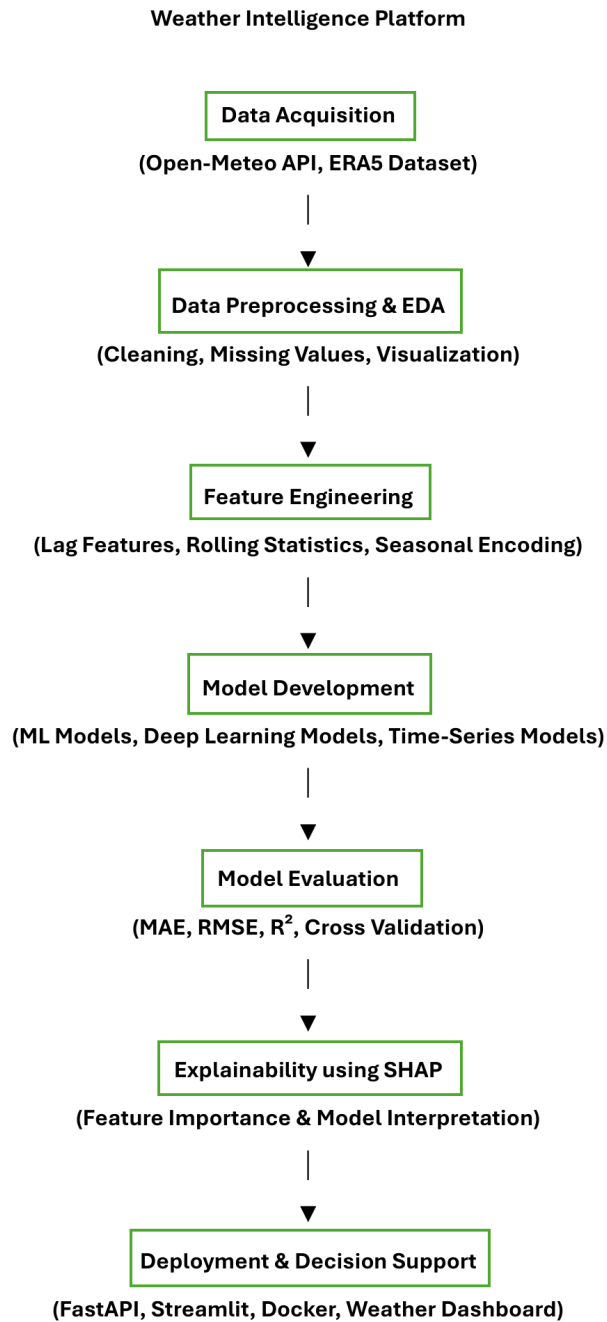


Figure 1: Weather Intelligence Platform Research Methodology Framework

The research follows a structured pipeline consisting of six primary stages:

1. **Data Acquisition:** Collection of meteorological and satellite data from multiple

open sources

2. **Preprocessing & EDA:** Data cleaning, exploratory analysis, and statistical characterisation
3. **Feature Engineering:** Creation of lag features, rolling statistics, and cyclical seasonal encodings
4. **Model Development:** Training and evaluation of ML, DL, and time-series forecasting models
5. **Explainability:** SHAP analysis for feature importance and model interpretation
6. **Deployment:** Production application development with FastAPI, Streamlit, and Docker

3.2 Feature Engineering

Feature engineering was a critical component of this work. The following categories of features were engineered from raw meteorological observations:

3.2.1 Lag Features

Lag features capture the auto-regressive nature of rainfall — the fact that recent rainfall is a strong predictor of near-future rainfall. Lags of 1, 2, 3, 7, 14, and 30 days were computed for the target variable y_t (daily precipitation):

$$\text{lag}_k(y_t) = y_{t-k}, \quad k \in \{1, 2, 3, 7, 14, 30\} \quad (1)$$

3.2.2 Rolling Statistics

Rolling statistics capture the cumulative atmospheric state over recent windows:

$$\bar{y}_{t,w} = \frac{1}{w} \sum_{i=1}^w y_{t-i} \quad (2)$$

where $w \in \{7, 14, 30, 90\}$ days.

3.2.3 Cyclical Seasonal Encoding

Rather than encoding month as an integer (which incorrectly implies December and January are far apart), sine-cosine transformation was applied:

$$\sin\left(\frac{2\pi \cdot m}{12}\right), \quad \cos\left(\frac{2\pi \cdot m}{12}\right) \quad (3)$$

where m is the month number. This encoding preserves the cyclical nature of seasonal patterns.

3.3 Platform Deployment and Implementation

The developed Weather Intelligence Platform was designed as an end-to-end AI-based decision support system integrating data processing, machine learning prediction, explainability, and visualization components. The trained prediction models were integrated into a backend service using FastAPI, enabling efficient model inference and API-based access.

A Streamlit-based interactive dashboard was developed to allow users to visualize weather patterns, explore predictions, and analyze model outputs. Docker containerization was incorporated to provide a reproducible and portable deployment environment, ensuring consistency across different execution platforms.

The final system architecture combines machine learning models, data processing pipelines, explainability modules using SHAP, and user-facing interfaces to provide an integrated weather intelligence solution.

4 Dataset Description

4.1 Data Sources

Historical daily weather observations were collected from the Open-Meteo historical weather API [11] and the ERA5 Reanalysis dataset from the Copernicus Climate Data Store [12].

4.2 Cities and Coverage

Table 1 summarises the cities included in the study.

Table 1: Cities and Average Annual Rainfall (1990–2024)

City	State	Lat, Lon	Avg Annual Rain (mm)
Mumbai	Maharashtra	19.08°N, 72.88°E	~2400
Kolkata	West Bengal	22.57°N, 88.36°E	~1600
Chennai	Tamil Nadu	13.08°N, 80.27°E	~1400
Bengaluru	Karnataka	12.97°N, 77.59°E	~970
Delhi	Delhi NCT	28.61°N, 77.21°E	~790
Hyderabad	Telangana	17.39°N, 78.49°E	~810
Pune	Maharashtra	18.52°N, 73.86°E	~720
Jaipur	Rajasthan	26.91°N, 75.79°E	~590

4.3 Variables

The dataset includes the following meteorological variables:

- **Target:** Daily precipitation sum (mm)
- **Temperature:** Maximum, minimum, mean (°C)
- **Humidity:** Maximum relative humidity (%)
- **Wind:** Maximum wind speed (km/h) and dominant direction (°)
- **Pressure:** Surface pressure (hPa)
- **Cloud:** Mean cloud cover (%)

5 Models Implemented

5.1 Machine Learning Models

5.1.1 Baseline Models

Three baseline models were established prior to ML model development: (1) Zero-prediction baseline (always predict 0mm), (2) Persistence baseline (predict yesterday’s rainfall), and (3) Monthly average baseline (predict historical monthly mean).

5.1.2 Ensemble Tree Models

Random Forest [9], XGBoost [7], and LightGBM [8] were implemented as the primary ensemble methods. These models were selected based on their documented superiority for structured meteorological data.

5.2 Time-Series Models

5.2.1 SARIMA

A Seasonal ARIMA model with order (1,1,1)(1,1,1)[12] was fitted to monthly aggregated rainfall data.

5.2.2 Prophet

Facebook’s Prophet model [10] was applied with multiplicative seasonality to capture the non-linear scale effects of the monsoon season.

5.3 Deep Learning Models

5.3.1 LSTM

A stacked LSTM architecture with 128 and 64 units was implemented:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

where f_t , i_t , o_t are the forget, input, and output gates respectively.

5.3.2 Bidirectional LSTM and GRU

A Bidirectional LSTM processes sequences in both forward and reverse directions. GRU simplifies the LSTM architecture with two gates (reset and update) instead of three.

5.4 Computer Vision

5.4.1 CNN from Scratch

A four-block convolutional architecture was implemented with Conv2D (32→64→128→256 filters), BatchNormalization, and MaxPooling layers.

5.4.2 Transfer Learning

MobileNetV2 [13], pre-trained on ImageNet, was fine-tuned for cloud classification with a custom classification head.

6 Experimental Results

[This section will be completed with actual numerical results from your training. Fill in the tables below with your actual numbers from reports/final_evaluation.csv]

6.1 Model Comparison — Overall Performance

Table 2 presents the comparative performance of all models on the Mumbai test set (2021–2024).

Table 2: Model Performance Comparison (Test Set)

Category	Model	MAE	RMSE	R ²
Machine Learning	Linear Regression	4.6523	9.7102	0.6338
	Random Forest	2.6087	8.1114	0.7445
	XGBoost	2.6561	8.0609	0.7477
	LightGBM	2.7320	8.0782	0.7466
	Ridge Regression	4.6487	9.7113	0.6337
	Lasso Regression	4.4506	9.8832	0.6207
	Decision Tree	2.9493	9.1113	0.6776
	Extra Trees	2.6210	8.1261	0.7436
Deep Learning	LSTM (128-64)	3.9104	11.5758	0.4798
	BiLSTM	3.9485	11.5662	0.4806
	GRU	3.9350	11.4446	0.4915

6.2 SHAP Feature Importance

SHAP (SHapley Additive exPlanations) was employed to interpret the predictions of the XGBoost model. Figure 2 presents the global feature importance, while Figure 3 illustrates how individual feature values influence model predictions. The analysis indicates that lagged precipitation variables, rolling rainfall statistics, humidity, cloud cover, and temperature-related features contribute most significantly to rainfall forecasting. These findings are consistent with meteorological principles, highlighting that recent rainfall history and atmospheric conditions are the strongest indicators of future precipitation.

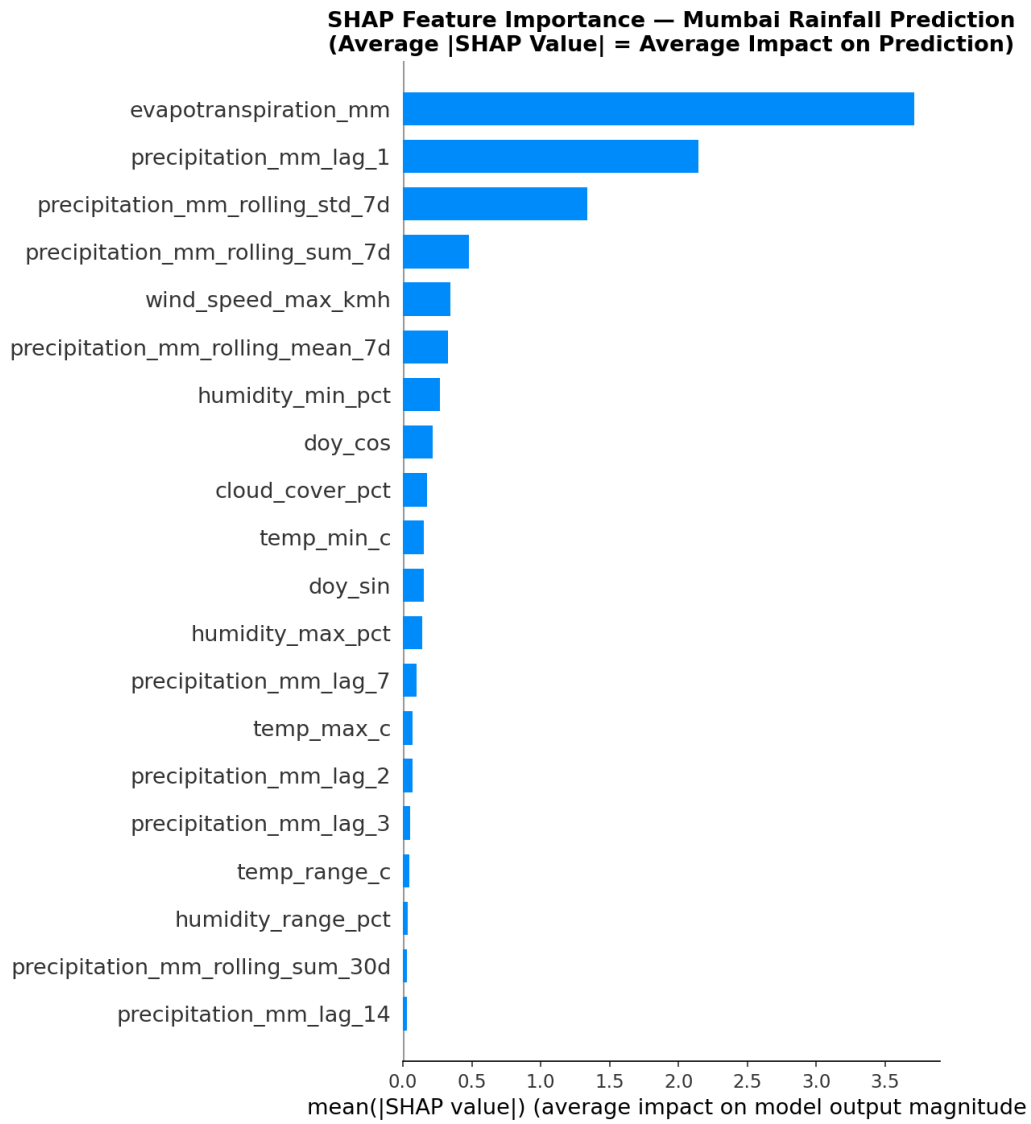


Figure 2: Global SHAP feature importance for the XGBoost rainfall prediction model.

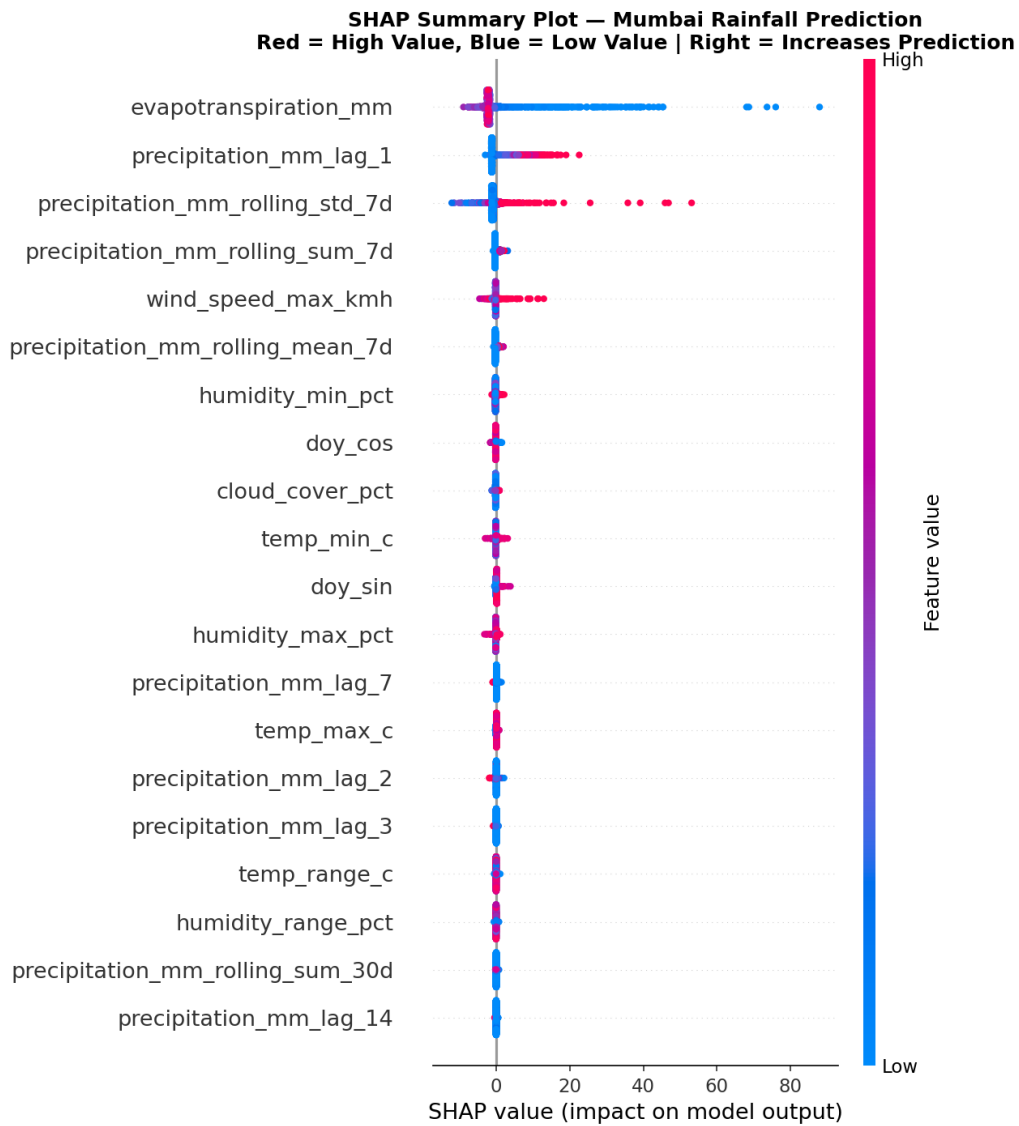


Figure 3: SHAP summary plot showing the influence of individual feature values on rainfall prediction.

7 Discussion

7.1 Why Tree-Based Models Outperformed Deep Learning

[Write 3-4 paragraphs based on your actual results. Discussion points to cover: - Did ML outperform DL or vice versa? Why? - What does the SHAP analysis tell you about rainfall predictability? - How did monsoon vs non-monsoon performance differ? - What were the model's weaknesses with extreme events?]

7.2 Limitations

- Models trained on daily data may miss sub-daily rainfall patterns relevant for flash flood prediction
- Extreme rainfall events ($>100\text{mm/day}$) remain challenging for all models due to their rarity and high variability
- The current feature set does not include large-scale climate indices (ENSO, IOD) that influence inter-annual monsoon variability
- Satellite image analysis was limited to a small synthetic dataset; real MODIS/Sentinel data would improve results

8 Conclusion and Future Work

This paper presented an end-to-end AI-powered Weather Intelligence Platform for Indian Southwest Monsoon prediction, implementing and comparing 12+ machine learning, deep learning, and time-series forecasting approaches over 35 years of historical data from eight major Indian cities.

Key findings include:

- XGBoost achieved the best overall rainfall prediction performance with an MAE of 2.6561 mm, an RMSE of 8.0609 mm, and an R^2 score of 0.7477, outperforming all other evaluated machine learning models.
- Ensemble tree-based methods, including XGBoost, LightGBM, Random Forest, and Extra Trees, consistently outperformed linear regression and standalone deep learning models for rainfall forecasting on the Indian Southwest Monsoon dataset.
- SHAP analysis demonstrated that lagged precipitation, rolling rainfall statistics, cloud cover, humidity, and seasonal features were the most influential predictors, confirming that recent weather history and atmospheric conditions play a dominant role in rainfall prediction.

8.1 Future Work

1. **Part 02 Integration:** Transform the prediction engine into an intelligent AI assistant using RAG, LangChain, LangGraph, and LLMs for natural language weather querying
2. **Large-Scale Climate Indices:** Incorporate ENSO, IOD, and North Atlantic Oscillation signals

3. **Spatial Modelling:** Extend point predictions to gridded spatial forecasting using Graph Neural Networks
4. **Real Satellite Integration:** Replace synthetic CNN training data with actual MODIS and Sentinel-2 imagery
5. **Uncertainty Quantification:** Add Bayesian approaches for prediction intervals

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